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UNIVERSITIES IN THE DIGITAL ERA AND THE INDUSTRIAL REVOLUTION 4.0
(Summary of Inaugural Public Lecture Material on September 24, 2025)

1. Challenges or Opportunities for Education Leadership in the Era of the Industrial Revolution 4.0.
   a. Industrial Revolution 4.0 Bang Automation Technology.
   b. Cyber War.
   c. Society 5.0
   d. Era Disruption.
   e. It was Postruth.
   f. Conflict of Interest.
   g. Vucah World: Volatility, Uncertainty, Complexity, Ambiguity, Hyper Conectivity.

   21st Century Challenges:
   a. Jaminan Mutu (Quality Control).
   b. Speed.
   c. Convinience.
   d. Agewave.
   e. Choice.
   f. Variety of Lifestyle.
   g. Price Competition (Discounting).
   h. Value Added.
   i. Customer Service.
   j. Technology as a Mainstay (Technoage).

   AI is the core technology of automation.

   Industrial Revolution 4.0.
   This phenomenon is predicted to trigger the presence of various trends in the future, including in the industrial sector.

   Increased automation.
   There will be a significant increase in automation in the business world, especially in workflow automation, Optical Character Recognition (OCR), and Robotic Process Automation (RPA).

   AI is a core technology at the logical layer in the application of industry 4.0.

   The world has entered the era of the 4th Industrial Revolution (IR 4.0), the impact of which is an increase in productivity and efficiency of the entire production value chain, thus affecting competitiveness at the global level.

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In the application of Industry 4.0, there are 5 technologies that are divided into 3 layers, namely:

a. Logical Layer: AI.
b. Connectivity Layer: IoT.
c. Physical Coating: AR/VR, Advanced Robotics and 3D Printing.

The government must come up with policies that can mitigate the negative impacts of AI such as job displacement where around 17% of jobs in Indonesia can be automated by AI.

The number of incidents related to AI abuse is increasing rapidly. The number of AI incidents and controversies has increased 26 times since 2012, as have fake videos.

Some uses of AI technology can be high-risk, so it can harm society, if not managed properly.
Examples of AI applications on:

a. Biometric identification in public spaces.
b. Autonomous weapons.
c. Driverless vehicles.
d. Diagnostics and medical treatment.
e. Credit score.
f. Resume Scanning.

AI needs to be navigated to ensure public safety by encouraging ethics and fair practices in its implementation.

2. Developments in World Policy on the Use of Digital and AI.

Several countries in the world already have regulations/policies related to the use of AI that are tailor-made for the needs of their countries.
The United States, the PRC and the European Union have different approaches to how to regulate AI, so each entity/country must have a vision and mission to regulate AI that meets the needs of its entity/country.

The PRC's priorities are on the projection of National Power, particularly military technology and digital security.
The AI Sovereignty approach is adopted by Russia.

The European Union's priority on regulation for the protection of privacy and user rights and digital security.
AI Regulation approach adopted by Japan, France, Germany

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The United States prioritizes innovation and R&D.
The AI Liberalization approach was adopted by Israel, South Korea, Singapore, Australia, Estonia.

Generative AI has the potential to add USD 243.5 billion or 18% of GDP in 2022.

Contribution of production capacity by sector:
a. Manufacturing.
b. Construction.
c. Wholesale and Retail Trade.
d. Transport Services.
e. Agriculture, Hunting, Forestry, and Fishing.

Indonesia is well-positioned to seize AI opportunities.
Indonesia is currently the largest AI market in ASEAN and the 4th in Asia.
Indonesian workers are optimistic that AI will have a positive impact on their careers. As many as 56% believe that AI will increase productivity and work efficiency.

3. Higher Education in the Era of the Industrial Revolution 4.0 and Society 5.0.

Higher Education is:
a. The culmination of qualifications, resource systems, and educational standards.
b. The determinant of change, improvement and progress of the nation and civilization.
c. Qualifications and relevance: graduates, research and innovation products are the main pillars and indicators of performance.
   Academic Competence (contentual) related and relevant to Work Competence (contextual).
   Educational Criteria and Standards related to and relevance to Professional Criteria and Standards.
   Accreditation of Education Programs related to and relevance to Professional Program Accreditation.

Creating an education system that is inclusive, world-class, and accessible to all, to improve the qualifications of the workforce in the digital era.

Intervention through 3 strategic imperatives:
a. Formal Education: Improving the quality of the basic education system (K-12) and higher education through the use of digital technology.
b. Workforce Empowerment: Establish inclusive digital education and training programs to prepare the workforce with the skills of the ICT sector's future.
c. Life-Long Learning: Creating a digitally literate society by utilizing digital devices for more productive use.

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Carried out by Universities in the Era of the Industrial Revolution 4.0 and Society 5.0.

a. Collaboration and Partnership.
b. Evaluation and Research.
c. Student Involvement.
d. Professional Development for Lecturers.
e. Flexible Curriculum.
f. Application of Digital Technology and AI.

Application of Digital Technology and AI:

a. Infrastructure: The availability of supporting infrastructure, such as internet networks, hardware and software.
b. Training and Education: The importance of training to ensure users can utilize technology effectively.
c. Policies and Regulations: A legal framework that facilitates or inhibits the use of technology.
d. Culture and Social Adoption: People's attitudes toward new technologies can affect how quickly and widely they are adopted.
e. Sustainable Innovation: The process of applying technology must be balanced with innovation to remain relevant to the needs of society.

Creating high-speed Internet connectivity that is accessible to all walks of life, reliable and affordable, thus supporting the adoption of a wide range of digital use-cases.

Intervention through 3 strategic imperatives:

a. Coverage: Build inclusive high-speed internet access through advanced connectivity solutions.
b. Quality: Developing high-performance, durable and efficient fiber optic networks through industry collaboration.
c. Computing Edge: Making Indonesia a regional computing hub that is able to accommodate the increase in data traffic.

Creating an education, a productive, advanced, high-value-added and globally harmonious business ecosystem by maximizing the use of smart digital technology.

Intervention through 4 strategic imperatives:

a. Digital MSMEs: Encourage the adoption of products and services by MSMEs to increase market access and business productivity.
b. Technology Start-ups: Promote the creation of new technology unicorns through the development of an attractive start-up ecosystem.
c. Key Sector Digital: Promote the use of innovative digital and smart technologies that aim to increase productivity and added value in priority sectors.
d. Cybersecurity: Increasing public awareness of system security and developing the quality of cybersecurity technology.

Enhance superior R&D capabilities in specific focus areas to accelerate digital transformation and spur innovation.

Intervention through 2 strategic imperatives:

a. Research and Development Commitment: Train the skilled talents necessary to develop research and development activities.
b. Innovation Culture: Encourage the development and commercialization of innovation across priority sectors to increase business productivity.

4. Ethics of the Use of Information Technology in the Higher Education Environment.

Ethical Principles:

a. Data Privacy and Security.
b. Access Fairness.
c. Copyright and License Compliance.
d. Transparency and Accountability.
e. Ethics in Teaching and Learning.
f. Social Responsibility/Use of Technology for Good.

Ethics of Creating an Educational Ecosystem:

a. Development of Ethics Policy with Socialization and Training.
b. Ethics Discussion Forum through Collaboration with Ethicists.
c. Monitoring the Use of Technology with Feedback from Users.
d. Integrating Ethics in the Curriculum with Extracurricular Activities.
e. Partnerships with Industry with Community Engagement.
f. Readiness to Face New Technologies Using Technology for Ethical Research.

Key Performance Indicators to drive transformation in higher education:

a. Quality of Graduates:
   Graduates get decent jobs.
   Students gain off-campus experience.
b. Quality of Lecturers and Lecturers:
   Lecturers carry out activities outside the campus.
   Practitioners teach on campus.
   The work of lecturers is used by the community and gets international recognition.
c. Curriculum Quality:
   The study program collaborates with world-class partners.
d. Study programs of international standards.